



Python Mastery Roadmap: From Zero to Logic

For: CSE AI/ML Student (3rd Year)

Goal: Master Core Python to prepare for DSA & AI

Method: Active Recall & Toy Projects (No Passive Watching!)



How to Use This Document

1. **Don't binge-watch.** Watch the specific topic video, then **IMMEDIATELY** build the Toy Project.
2. **The "Delete Rule":** If you copy code to make it work, delete it and write it again from memory.
3. **Use the "AI Prompt" (Page 3)** whenever you start a new chat session.



Phase 1: The Building Blocks (Days 1-4)

Focus: Storing data and Basic Math.

Topics to Cover

1. **Variables & Data Types:** int, float, str, bool.
2. **Input/Output:** print(), input(), f-strings.
3. **Type Casting:** Converting str to int (e.g., int("5")).
4. **Operators:** +, -, *, /, % (Modulus), ** (Power).



Best Resources (Telugu)

- **Watch (Telugu):** [Vamsi Bhavani - Python Full Course \(Watch first 2 Hours\)](#)
 - *Note: This is a 10-hour video. For Phase 1, just watch the "Variables", "Data Types", and "Input/Output" sections.*
- **Watch (Telugu Alternative):** [Python Life - Python Full Course](#)
- **Read:** [GeeksforGeeks - Python Basics](#)



Toy Projects (Do NOT Move Forward Until These Are Done)

- **Project 1: The Greeter**
 - *Task:* Ask user for Name & Birth Year. Print: "Hello [Name], you are [Age] years old."
- **Project 2: The Tip Calculator**
 - *Task:* Ask for Bill Amount (e.g., 1000) and Tip Percent (e.g., 10). Print total amount.



Phase 2: The Logic (Days 5-9) — CRITICAL

Focus: Making decisions and repeating tasks.

Topics to Cover

1. **Conditionals:** if, elif, else, Nested ifs.
2. **Logical Operators:** and, or, not.
3. **Loops:** for loop (range), while loop.
4. **Control:** break, continue.

Best Resources (Telugu)

- **Watch (Telugu):** [Vamsi Bhavani - Control Statements \(Look for 'If Else' and 'Loops' chapters\)](#)
- **Practice:** [GeeksforGeeks - Control Flow](#)

Toy Projects

- **Project 3: Traffic Cop**
 - *Task:* Ask for speed. If > 80 print "Fine". If > 60 print "Warning". Else print "Safe".
- **Project 4: Multiplication Table**
 - *Task:* Ask for a number (e.g., 5). Use a loop to print the table (5x1... 5x10).
- **Project 5: Guess the Number Game**
 - *Task:* Hardcode a secret number (e.g., 42). Let user guess repeatedly using a while loop until they get it right.



Phase 3: Data Organization (Days 10-12)

Focus: Managing lists of data.

Topics to Cover

1. **Lists:** Indexing [0], Slicing [1:5], Methods (.append, .pop, .sort).
2. **Strings:** Methods (.upper, .lower, .replace, .split).
3. **Dictionaries:** Key-Value pairs (Basics only).

Best Resources (Telugu)

- **Watch (Telugu):** [Python Life - Lists and Strings](#)
- **Read:** [GeeksforGeeks - Python Data Structures](#)

Toy Projects

- **Project 6: Shopping Cart**
 - *Task:* Create an empty list. Use a loop to let user "Add" items. Print the full list at the end.
- **Project 7: Vowel Counter**
 - *Task:* Ask for a sentence. Loop through it and count how many vowels (a,e,i,o,u) are in

it.



Phase 4: Reusability (Days 13-14)

Focus: Clean Code.

Topics to Cover

1. **Functions:** def, Parameters, Return.
2. **Scope:** Global vs Local variables.



Toy Projects

- **Project 8: The Math Wizard**
 - *Task:* Create functions add(a,b), sub(a,b), mul(a,b). Ask user which operation they want and call the right function.



The "New Chat" Prompt

Copy the text below and paste it when you start a new AI chat. It will force the AI to become your strict mentor.

COPY FROM HERE DOWN:

"Hi! I am a 3rd Year CSE AI/ML student working on mastering **Core Python**.

My Current Status:

I am following a 'Toy Project' roadmap. I focus on Logic Building over memorizing syntax.

My Rules for You:

1. **Do NOT give me full code solutions.** If I am stuck, give me a HINT or ask me a guiding question.
2. **Explain like I am a beginner.** Use real-world analogies (like Kitchen, Traffic, Shopping).
3. **Verify my understanding.** After I write code, ask me to explain *why* it works.

Where I am right now:

I am currently on [INSERT TOPIC HERE, e.g., Phase 2 Loops].

Please give me a small 'Toy Project' challenge for this topic to test my skills."